

Antibody–Antigen Affinity Prediction from Amino Acid Sequences using Protein Language Models and Generative Data Augmentation

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Abstract

Predicting antibody–antigen binding affinity from amino acid sequences is a challenging problem due to limited experimental data and the complexity of molecular interactions. While structure-based approaches achieve strong performance, they require resolved 3D structures and limit scalability. We propose a sequence-based regression framework leveraging pretrained protein language models to encode heavy chain, light chain, and antigen sequences with higher-order interaction modeling. We introduce a structure-guided generative augmentation pipeline that expands the training set by two orders of magnitude while preserving structural consistency. Generated samples inherit affinity labels and include generative confidence scores.

1 Introduction

Protein interactions are fundamental to biological function, governing processes such as immune recognition, signal transduction, and molecular regulation. In particular, antibody–antigen binding is driven by biochemical compatibility and physical interactions between molecular surfaces.

The biological process underlying protein function follows a sequence-to-structure paradigm: amino acid sequences fold into three-dimensional (3D) structures that determine their functional properties. This folding process is governed by physical forces such as hydrogen bonding, electrostatic interactions, hydrophobic effects, and van der Waals forces [10, 7]. Recent advances in deep learning have significantly improved protein structure prediction, most notably with models such as AlphaFold [13] [3] and related large-scale structure prediction frameworks.

Binding affinity between molecules is commonly quantified by the dissociation constant K_d , which measures the equilibrium strength of molecular interactions:

$$K_d = \frac{[A][B]}{[AB]}$$

A lower K_d indicates stronger binding affinity. Importantly, K_d reflects the underlying free energy landscape of molecular interactions and is therefore directly linked to structural complementarity and physical interaction energies.

Several recent approaches have attempted to predict binding affinity from 3D structural information. Structure-based deep learning models leveraging geometric deep learning, graph neural networks, and equivariant neural networks have been proposed to estimate binding affinity directly from atomic coordinates [8, 17]. These methods typically operate on resolved or predicted 3D structures and model spatial interactions explicitly.

However, access to reliable 3D structures remains limited and computationally expensive. In contrast, predicting binding affinity directly from amino acid sequences removes the dependency on explicit structure reconstruction and enables scalable training on large biological datasets.

In this work, we investigate whether antibody–antigen affinity can be accurately predicted directly from sequence representations. By leveraging protein language model embeddings combined with generative data augmentation, we aim to approximate the physical interaction mechanisms governing molecular binding without requiring explicit 3D structural input.

2 Data

2.1 Dataset Description

Our experiments are conducted on the Structural Antibody Database (SAbDab), a curated database of experimentally resolved antibody–antigen complex structures with associated biochemical annotations [11] [2].

SAbDab collects antibody–antigen complexes from the Protein Data Bank (PDB) and provides standardized annotations including antibody heavy chain (H-chain) sequences, light chain (L-chain) sequences, antigen sequences, and experimentally measured dissociation constants K_d when available. After filtering for complete sequences and available K_d values, our final dataset contains approximately 600 antibody–antigen complexes.

For each complex in our study, we extract:

- Heavy chain amino acid sequence,
- Light chain amino acid sequence,
- Antigen amino acid sequence,
- Experimentally measured dissociation constant K_d .

Only complexes with complete sequence information and available affinity measurements are retained for training and evaluation.

2.2 Affinity Representation

Since dissociation constants span several orders of magnitude and exhibit strong skewness, we transform the affinity values into the negative logarithmic scale:

$$y = -\log_{10}(K_d)$$

This transformation improves numerical stability and is standard practice in binding affinity prediction tasks, as it converts raw thermodynamic constants into a more interpretable and model-friendly representation.

2.3 Distribution Characteristics and Bias

We analyze the empirical distribution of $-\log_{10}(K_d)$ across the dataset. The distribution is highly non-uniform and concentrated in specific affinity ranges, reflecting experimental measurement preferences and reporting biases in the literature. Such imbalance may introduce training bias, where models overfit to overrepresented affinity regimes

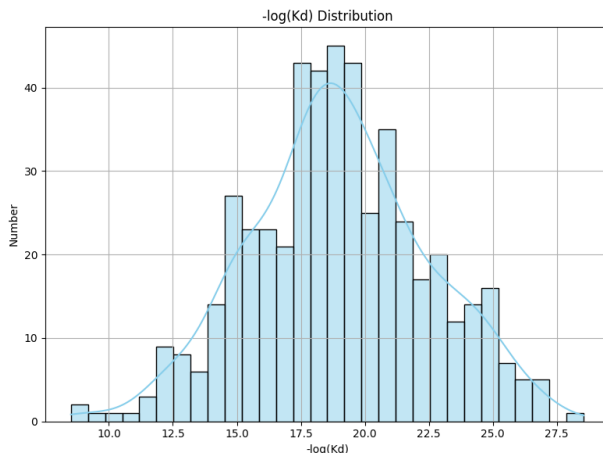


Figure 1: Distribution of SabDab dataset

and underperform on underrepresented regions. Understanding this distribution is therefore critical for designing appropriate training strategies and data augmentation procedures.

3 Data Augmentation

3.1 Motivation

The limited number of experimentally characterized antibody–antigen complexes significantly restricts the training capacity of deep learning models. After filtering SAbDab for complete sequence information and reliable affinity measurements, the number of usable complexes remains on the order of a few hundred, which is insufficient for training high-capacity sequence-based models.

To mitigate this limitation, we design a structure-guided generative data augmentation strategy that expands the dataset while preserving biophysical consistency.

3.2 Structure-Guided Generation Pipeline

For each antibody–antigen complex in the original dataset, we first retrieve the corresponding structural information from the Protein Data Bank (PDB) unit files. This allows us to associate each amino acid sequence triplet (heavy chain, light chain, antigen) with its experimentally resolved three-dimensional structure.

Starting from these 3D structures, we apply generative protein models such as AlphaFold [13] [3] -based refinement models, Boltz-2 [16] [6] , and ProteinMPNN [9] [4] to generate novel amino acid sequences that are structurally compatible with the original complex.

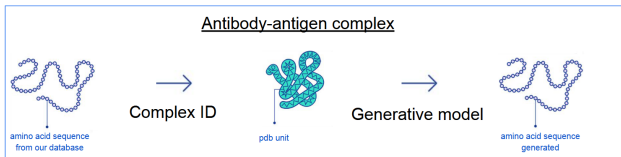


Figure 2: Data Augmentation Pipeline

The generative models propose alternative amino acid sequences that preserve structural compatibility with the given 3D conformation.

3.3 Affinity Label Propagation

A key aspect of our augmentation strategy is that the generated sequences inherit the dissociation constant K_d from the original complex.

Since the generation process is conditioned on a structure associated with an experimentally measured affinity, we assume that the newly generated sequences remain in the same affinity regime as the original complex.

Therefore, each synthetic sample is assigned:

- The inherited K_d value from the source complex,
- The corresponding transformed label $y = -\log_{10}(K_d)$,
- A generation confidence score provided by the generative model.

3.4 Use of Generation Confidence Scores

An important component of our augmentation strategy is the retention of the confidence score produced by the generative model.

This score reflects how well the generated amino acid sequence matches the structural constraints imposed during generation. Rather than discarding this information, we explicitly store it and use it for downstream filtering and potential sample weighting during training.

In total, this procedure increases the number of training samples from approximately 500 experimentally measured complexes to around 50,000 augmented samples, representing a 100-fold expansion of the dataset.

3.5 Analysis of Generation Scores and Affinity Correlation

A natural hypothesis underlying our generative augmentation strategy is that the confidence score produced by the generative model could correlate with the binding affinity K_d of the underlying complex.

Intuitively, we would expect that complexes characterized by stronger binding affinity (low K_d , high $-\log_{10}(K_d)$) would correspond to more structurally stable and physically consistent configurations. If this assumption held, generation scores could serve as a proxy for affinity quality and potentially guide sample weighting during training.

To evaluate this hypothesis, we computed the Pearson correlation between:

- The generation scores produced by different generative pipelines (ProteinMPNN [9] [4] , fine-tuned ProteinMPNN [12], Rosetta [14] [1]-based refinement, and other structure-conditioned methods),
- The associated transformed affinity labels $y = -\log_{10}(K_d)$.

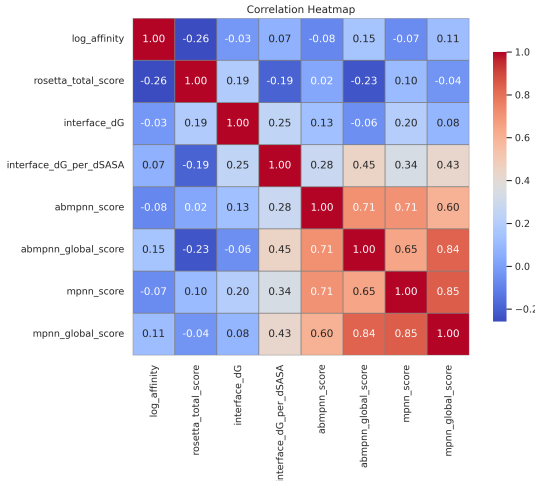


Figure 3: Spearman Correlation Heatmap

Figure 3 shows the correlation matrix between affinity and various structural and generative scores.

Empirically, we observe no strong correlation between generation scores and binding affinity values. While generative scores exhibit strong mutual correlations across similar model variants, their correlation with $-\log_{10}(K_d)$ remains weak.

This result suggests that generative confidence scores primarily measure structural plausibility under the generative model prior rather than biological binding strength. In practice, this implies that generation scores cannot be directly used as a reliable proxy for affinity and should instead be interpreted as structural quality indicators.

We therefore retain generation scores as auxiliary information for potential downstream filtering or weighting, but do not assume any predictive relationship with K_d .

4 Method

4.1 Problem Formulation

Each antibody–antigen complex is composed of three biological sequences:

- Heavy chain sequence S_H
- Light chain sequence S_L
- Antigen sequence S_A

Given these three amino acid sequences, our objective is to predict the binding affinity expressed as:

$$y = -\log_{10}(K_d)$$

We model the problem as a supervised regression task:

$$f(S_H, S_L, S_A) \rightarrow \hat{y}$$

4.2 Sequence Embedding

Each sequence is independently encoded using a pretrained protein language model such as ESM-2 [15] [5] or ProteinMPNN [9] [4].

Let:

$$\phi_H = \text{Encoder}(S_H) \in \mathbb{R}^d$$

$$\phi_L = \text{Encoder}(S_L) \in \mathbb{R}^d$$

$$\phi_A = \text{Encoder}(S_A) \in \mathbb{R}^d$$

where the encoder produces a fixed-dimensional embedding for each biological component.

The three embeddings are computed independently and capture contextual structural and biochemical information encoded in the pretrained protein model.

4.3 Regression Model

The three embeddings are then combined to predict the binding affinity.

4.3.1 Multi-Layer Perceptron Architecture

We consider a regression model composed of three independent MLP branches:

$$h_H = \text{MLP}_H(\phi_H), \quad h_L = \text{MLP}_L(\phi_L), \quad h_A = \text{MLP}_A(\phi_A)$$

Each branch produces a scalar or low-dimensional representation.

The final affinity prediction is computed as:

$$\hat{y} = \sum_{i=1}^m h_H^{(i)} \cdot h_L^{(i)} \cdot h_A^{(i)}$$

where m denotes the output dimension of each branch.

This formulation explicitly models higher-order interactions between heavy chain, light chain, and antigen representations through multiplicative coupling.

Alternatively, we also experiment with a gradient boosting regression model such as XGBoost:

$$\hat{y} = \text{XGBoost}(\phi_H \oplus \phi_L \oplus \phi_A)$$

where $\oplus$ denotes concatenation of the three embeddings.

4.4 Training Strategy

The regression model is trained using supervised learning with a mean squared error loss:

$$\mathcal{L} = \frac{1}{N} \sum_{j=1}^N (\hat{y}_j - y_j)^2$$

Optimization is performed using the Adam optimizer with backpropagation.

We adopt a two-stage training strategy:

1. **Pretraining on Augmented Data:** The model is first trained on the augmented dataset (50,000 samples) to learn a robust initialization from synthetic diversity.

During this stage, we experiment with two different learning objectives:

- **Affinity Prediction Objective:** The model is trained to predict the inherited affinity label $y = -\log_{10}(K_d)$ associated with the original complex from which the augmented sequences were generated. In this setting, synthetic samples inherit the experimentally measured K_d of their source complex, and the model learns to regress directly on these propagated affinity values.
- **Generation Confidence Prediction Objective:** Instead of predicting affinity, the model is trained to predict the generation confidence score provided by the generative model during sequence sampling. This encourages the model to learn structural plausibility patterns captured by the generative process. The intuition is that the confidence score encodes structural consistency information that may provide useful representation bias even if it is not directly correlated with binding affinity.

2. **Fine-tuning:** The pretrained model is then fine-tuned on the original SAbDab dataset containing only experimentally measured complexes, in order to reduce potential bias introduced by synthetic data and to re-anchor the model on reliable affinity measurements.

This two-stage strategy allows us to compare the impact of affinity-supervised pretraining versus structure-confidence pretraining, and to analyze which surrogate objective provides better transfer to the downstream affinity prediction task.

5 Results

5.1 Quantitative Analysis

Table 1 reports the performance under different training strategies.

Training Strategy	Pearson	Spearman
No Data Augmentation	0.43	0.44
Augmentation (Inherited K_d)	0.53	0.56
Augmentation (Confidence Score)	0.58	0.62

Table 1: Performance comparison under different training strategies on the test set.

Impact of Data Augmentation We observe a consistent improvement in both Pearson and Spearman correlations when incorporating generative data augmentation.

Both augmentation strategies outperform the baseline trained without synthetic data, demonstrating the benefit of enriching the training distribution with generated samples.

Notably, pretraining with the generation confidence score achieves the best performance. This suggests that structural plausibility information captured by the generative model provides a stronger and more informative training signal than directly inheriting the original K_d values.

5.2 Qualitative Analysis and Prediction Bias

Figure 4 shows predicted versus ground-truth affinities under different training strategies.

The prediction plots reveal a systematic bias in the estimated affinity distribution.

Due to the highly non-uniform distribution of $-\log_{10}(K_d)$ in the training data, the model tends to regress toward the empirical mean. This leads to:

- Underestimation of extreme affinity values,
- Compression of predictions around high-density regions,

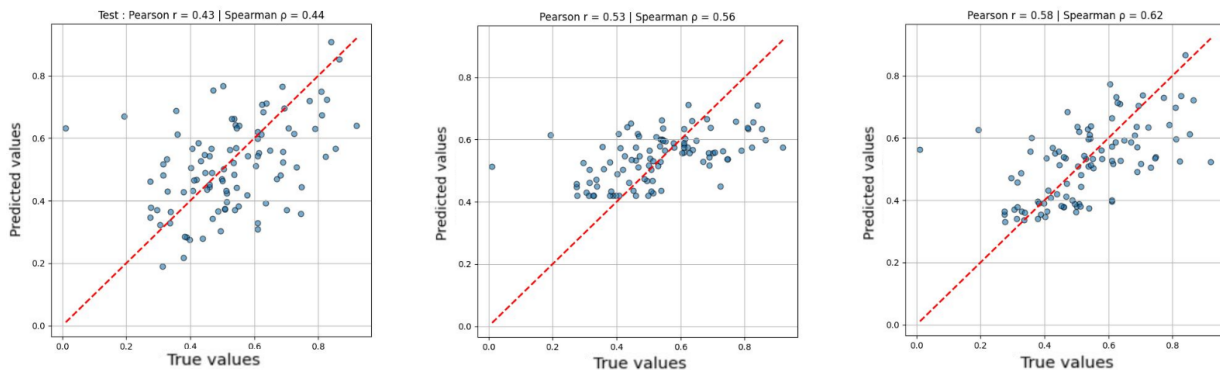


Figure 4: Predicted vs. ground-truth affinity under different training strategies. Left: no augmentation. Middle: augmentation with inherited K_d . Right: augmentation with confidence score.

- Reduced variance in the predicted distribution compared to the ground truth.

This behaviour confirms that the dataset is imbalanced in terms of affinity values and highlights the limitation of standard MSE training under skewed label distributions.

Data augmentation partially mitigates this effect by enriching the training distribution with synthetic samples. However, it does not fully remove the regression-to-the-mean phenomenon.

More advanced strategies such as distribution reweighting, quantile-aware losses, or loss functions robust to label imbalance could further reduce this bias.

6 Conclusion

Predicting antibody–antigen binding affinity directly from amino acid sequences remains a highly challenging task due to the limited availability of high-quality experimental data and the complex biophysical mechanisms underlying molecular interactions.

Our results demonstrate that generative data augmentation significantly improves predictive performance compared to training solely on experimentally measured complexes. In particular, leveraging structure-conditioned sequence generation and exploiting generative confidence signals provide meaningful gains in correlation metrics, indicating that synthetic diversity can help mitigate data scarcity.

However, despite these improvements, model performance remains far from saturation and prediction errors persist, especially in regions of sparse affinity coverage. This suggests that sequence-only representations combined with standard regression architectures are still limited in their ability to fully capture the underlying physical interactions.

To further improve performance, future work should focus on:

- Designing more expressive and physically-informed model architectures, and fine-tuning them directly on the augmented dataset,
- Exploring alternative generative pipelines for data augmentation to increase diversity,
- Investigating alternative regression targets derived from generative models or structural scores to better exploit synthetic supervision.

Overall, our study highlights both the potential and the limitations of generative augmentation for antibody–antigen affinity prediction, and opens directions for integrating stronger priors and more advanced training strategies.

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